BOARD OF INTERMEDIATE EDUCATION, KARACHI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

BOTANY

SECTION 'A' (Multiple Choice Questions) (16)

> Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed. Question (xv)'s options and question (xvi) were recovered from the top of the subjective scan page.)

> Note: TIME: 2 Hours — (40 Marks) — BOTANY 2022

Q. SECTION 'A' (Multiple Choice Questions) (16)

Q. 1 Choose the correct answer for each from the given options:

Qi. Enzyme are protected at high temperature by:

(A) Ice Crystal
(B) Super Cooling
(C) Freezing Tolera
(D) Heat Shock Proteins

Qii. Chromosomal Theory of Heredity was proposed by:

(A) Karl Correns
(B) Hershey and Chase
(C) George Beadle
(D) Walter Sutton

Qiii. In this sub-stage of Meiosis Prophase I Terminalization occurs:

(A) Pachytene
(B) Leptotene
(C) Diplotene
(D) Diakinesis

Qiv. A cell division without spindle is:

(A) Amitosis
(B) Mitosis
(C) Meiosis
(D) Apomixis

Qv. This is present in RNA but not in DNA.

(A) Uracil
(B) Cytosine
(C) Adenine
(D) Guanine

Qvi. The multiple effects of a single gene is termed as:

(A) Polygenic inheritante
(B) Multiple Alleles
(C) Pleiotropy
(D) Incomplete Dominance

Qvii. If the sequence of base in one chain is ATTGCAT, the base sequence of its Partner in the duplex is:

(A) ATAGCAT
(B) AATGCAT
(C) GCCATTA
(D) TAACGTA

Qviii. The first line of defence against pathogen in plants is:

(A) Endodemis
(B) Cortex
(C) Epidermis
(D) Cuticle

Qix. Gemination of seed within the fruit while still attached to the plant is:

(A) Hypogeal
(B) Viviparous
(C) Epigeal
(D) Oviparous

Qx. Promotion of flowering by cold treatment given to imbibed seeds is called:

(A) Vernalization
(B) Apomixis
(C) Parthenocarpy
(D) Phototropism

Qxi. Both polynucleotide strands of DNA remain separated by:

(A) 5A
(B) 10 A
(C) 15A
(D) 20A

Qxii. Dark zone of oceanic region is:

(A) Euphotic
(B) Aphotic
(C) Neritic
(D) Profundal

Qxiii. The cotyledon in monocot seed is called:

(A) Epicotyl
(B) Coleorhiza
(C) Coleoptile
(D) Scutellum

Qxiv. Example of ecosystem having lotic water is:

(A) Waterfall
(B) Swamp
(C) Pond
(D) Lake

Qxv. In guttation plants lose excess water in the form of liquid from special pores on leaves called:

(A) Sunken Stomata
(B) Hydathodes
(C) Lenticels
(D) Air pores

Qxvi. Death of cell due to injury is called:

(A) Apoptosis
(B) Nectosis
(C) Mitosis
(D) Meiosis

SECTION 'B' + Section C (Subjective)

Q. SECTION "B" (Short Answer Question) Marks: 16

> Note: 2. Attempt any Eight part questions. Each question carry one mark. Answer should not exceed two lines.

> Note: 3. Attempt any Four part questions Each question carries two marks. Answer should not exceed

Q. Section C (Detailed -Answer Question) (8

> Note: NOTE: Attempt any One question from this section. All questions carry equal marks.

ZOOLOGY

SECTION 'A' (Multiple Choice Questions) (18)

> Note: (Options in the original are separated by star marks, shown here as A–D in printed order. Check marks on some options in the scan are a previous solver's marks and are not transcribed.)

> Note: TIME: 2 Hours — (40 Marks) — ZOOLOGY 2022

Q. SECTION 'A' (Multiple Choice Questions) (18)

Q. 1 Choose the correct answer for each from the given options:

Qi. It is not a way of obtaining heat by organism:

(A) Solar Radiations
(B) Metabolism
(C) Muscle Contraction
(D) Sweating

Qii. The exoskeleton in cockroach is made up of:

(A) Bones
(B) Cartilages
(C) Chitin
(D) Calcium

Qiii. Parthenogenesis requires:

(A) one parent
(B) two gametes
(C) two parents
(D) no parent

Qiv. Contraction time in human eye muscle is:

(A) 0.001 second
(B) 0.01 second
(C) 0.1 second
(D) 1second

Qv. Cytoplasmic localization is consequence of:

(A) Fertilization
(B) Cleavage
(C) Morula
(D) Blastula

Qvi. The approximate number of human genes is:

(A) 10,000 to 15,000
(B) 20,000 to 25,000
(C) 30,000 to 35,000
(D) 40,000 to 45,000

Qvii. Number of tail vertebrae in Archaeopteryx were:

(A) 12
(B) 15
(C) 18
(D) 20

Qviii. Night blindness is due to the deficiency of:

(A) Vitamin K
(B) Vitamin A
(C) Vitamin C
(D) Minerals

Qix. These animals retain urea in their body to maintain hypersonic osmotic pressure:

(A) Mackerel Fish
(B) Trout Fish
(C) Sharks and Rays
(D) Amphibians

Qx. It is a bone of pectoral girdle:

(A) Ilium
(B) Ischium
(C) Pubis
(D) Scapula

Qxi. Deficiency of this neurotransmitter causes Parkinson's disease:

(A) Dopamine
(B) Acetylcholine
(C) Epinephrine
(D) Nor-Epinephrine

Qxii. The end of fertility in human female is called:

(A) Menarch
(B) Menses
(C) Menopause
(D) Ovulation

Qxiii. The sympathetic nervous system in humans consists of:

(A) Thoracic and Lumber Region
(B) Cranial Region
(C) Sacral Region
(D) Thoracic and Sacral Region

Qxiv. It is a multifactorial defect:

(A) Asthma
(B) Cystic Fibrosis
(C) Hungtinton's Disease
(D) Down's Syndrome

Qxv. These are postulates of Lamarck's theory of evolution except:

(A) Natural selection
(B) Inheritance of acquired character
(C) Effect of environment
(D) Use and disuse of organ

Qxvi. This hormone requires a second messenger.

(A) Testosterone
(B) Insulin
(C) Estrogen
(D) Progesterone

Qxvii. The study, of abnormal development is called:

(A) Teratology
(B) Ethology
(C) Gerontology
(D) Genetics

Qxviii. Sound higher than this level is called noise.

(A) 80 decibels
(B) 30 decibels
(C) 10 decibels
(D) 50 decibels

SECTION 'B' + Section C (Subjective)

Q. SECTION "B" (Short Answer Question) Marks: 18

> Note: NOTE: Attempt any NINE part questions in all. Select FIVE Reasoning Questions and FOUR Non-Reasoning Questions. All part questions carry equal marks.

Q. Section C (Detailed -Answer Question) (9)

> Note: NOTE: Attempt any ONE question from this section. All questions carry equal marks.